RobustGaSP: MLE vs MAP estimation

37D real data

AOD vs AOD

1 Variable 1: AOD at longitude -171.5625 and latitude 51.875

We start with AOD at the gridbox highlighted in green in Figure 1.

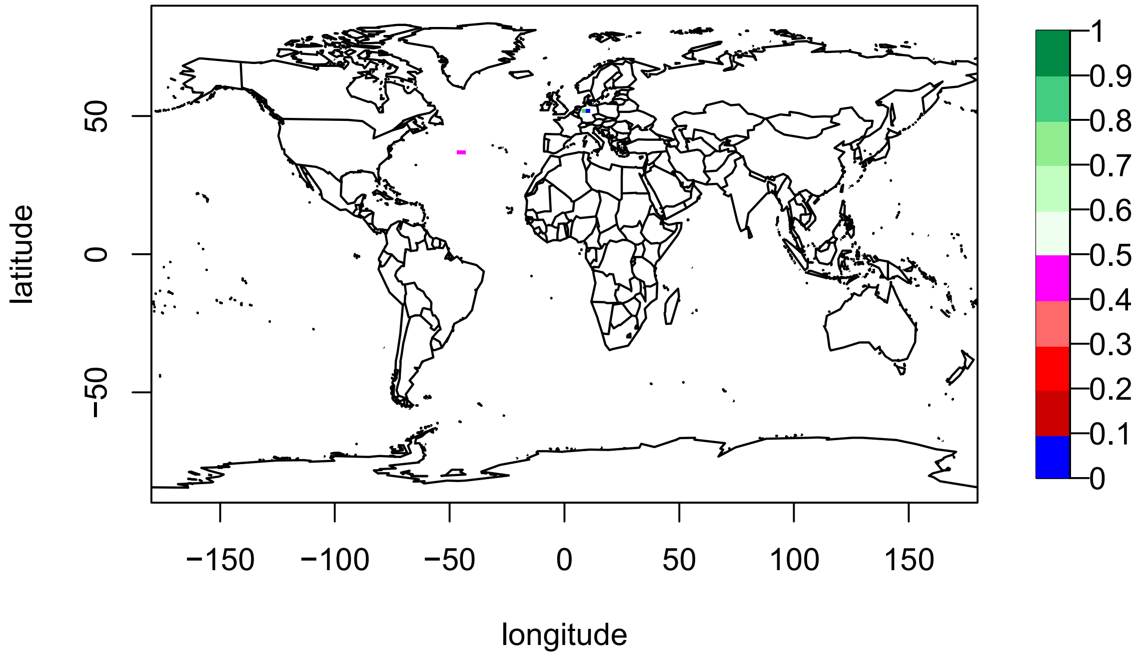
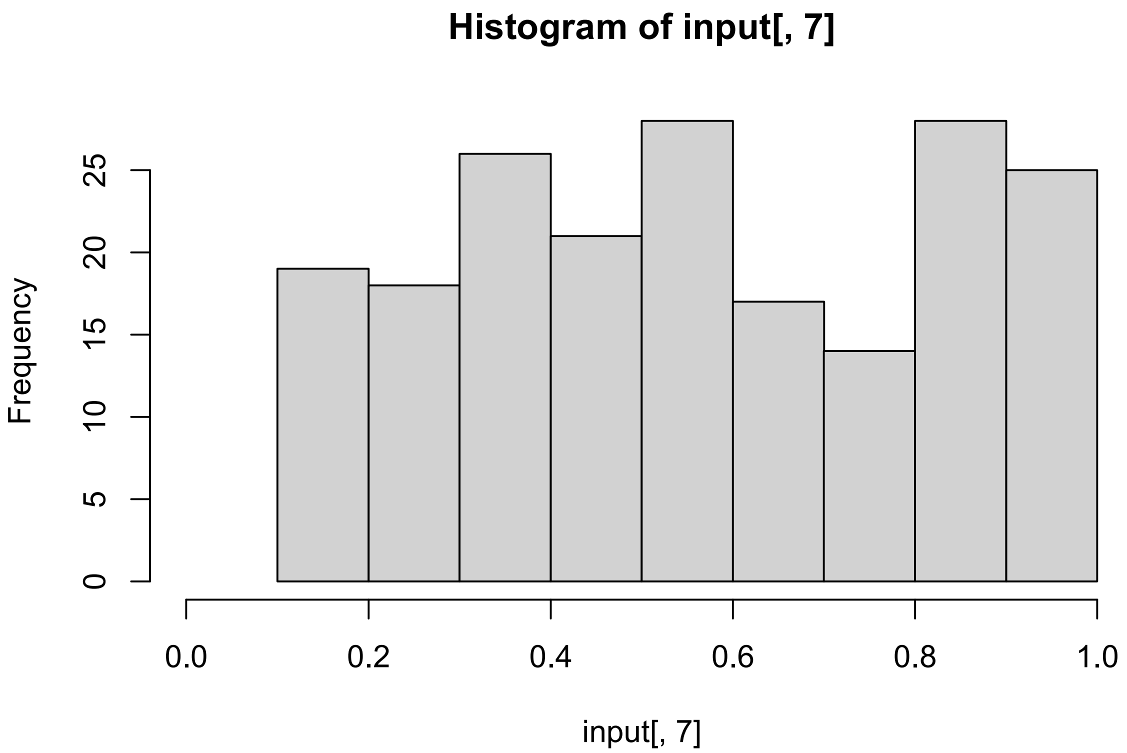


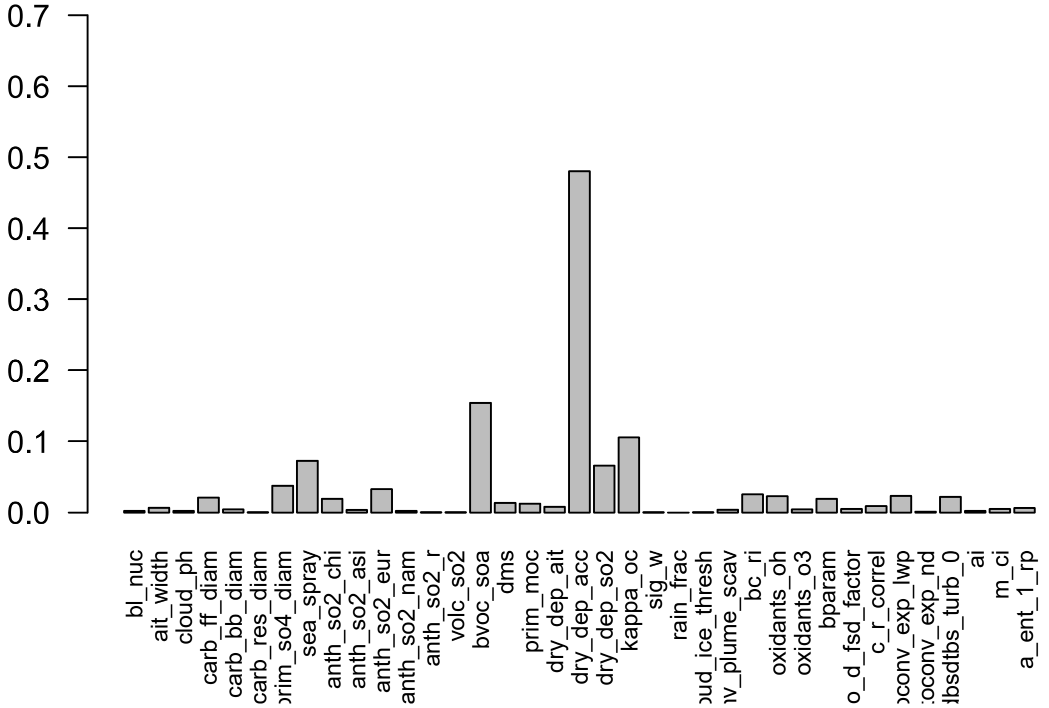
Figure 1: Selected gridbox is highlighted in green

1.1 Emulator fit and validation

SA gets done.

```
## [1] 196
```





```
## [1] "prim_so4_diam" "sea_spray"     "anth_so2_eur"  "bvoc_soa"
## [5] "dry_dep_acc"   "dry_dep_so2"   "kappa_oc"
```

```
## [1] 0.9490868
```

```r
input <- input[,which(mainEffects >= th)]
```

We have a 37-dimensional input, one output (AOD in July at the gridbox with longitude -171.5625 and latitude 51.875), and 196 noise-free observations. We will assume a linear mean function of the form

$$m(x) = \beta_0 + \beta_1(\text{bl\_nuc})\, x_1 + \beta_2(\text{ait\_width}) + \cdots + \beta_{37}(\text{a\_ent\_1\_rp})$$

The Gauss (squared exponential) *and* Matern 5/2 kernels will be compared for both estimation methods.

100 new LH points are generated and predictions at these points made. The mean width of the 95% CI's for these predictions are shown in Table 1.

Table 1: Mean width of 95% CI

Estimate method	Kernel	Mean width of 95% CI
MLE	Gauss	0.1768
MLE	Matern 5/2	0.1768
MAP	Gauss	0.1689
MAP	Matern 5/2	0.1853

Leave-one-out validation is carried out; the results are shown in Figure 2.

```
## The upper bounds of the range parameters are Inf Inf Inf Inf Inf Inf Inf
## The initial values of range parameters are 0.01381671 0.01436526 0.01435065 0.01435916 0.0143668 0.0
## Start of the optimization  1  :
## The number of iterations is  1
##  The value of the  profile likelihood  function is  90.17953
##  Optimized range parameters are 0.01381671 0.01436526 0.01435065 0.01435916 0.0143668 0.01435145 0.0
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 2.760073 3.084807 3.075348 3.080849 3.085803 3.075865 3.0
## Start of the optimization  2  :
## The number of iterations is  2
##  The value of the  profile likelihood  function is  90.17953
##  Optimized range parameters are 0.000000000000007141561 0.00001270957 0.00002029385 0.00000000000036
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The upper bounds of the range parameters are Inf Inf Inf Inf Inf Inf Inf
## The initial values of range parameters are 0.0138241 0.01436526 0.01435065 0.01435916 0.0143668 0.01
## Start of the optimization  1  :
## The number of iterations is  1
##  The value of the  profile likelihood  function is  89.71931
##  Optimized range parameters are 0.0138241 0.01436526 0.01435065 0.01435916 0.0143668 0.01435145 0.01
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 2.764068 3.084807 3.075348 3.080849 3.085803 3.075865 3.0
## Start of the optimization  2  :
## The number of iterations is  2
##  The value of the  profile likelihood  function is  89.71931
##  Optimized range parameters are 0.0000000000001298661 0.000000002391894 0.0000000001409969 0.000000
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The upper bounds of the range parameters are Inf Inf Inf Inf Inf Inf Inf
```

```
## The initial values of range parameters are 0.0138241 0.01436526 0.01435065 0.01435916 0.0143668 0.01
## Start of the optimization  1  :
## The number of iterations is  31
##  The value of the  marginal posterior  function is  155.9061
##  Optimized range parameters are 585.7556 1241383291861770532984008860862080428400484828 8 97660214246
##  Optimized nugget parameter is 0
##  Convergence:  FALSE
## The initial values of range parameters are 2.764068 3.084807 3.075348 3.080849 3.085803 3.075865 3.0
## Start of the optimization  2  :
## The number of iterations is  55
##  The value of the  marginal posterior  function is  65.97625
##  Optimized range parameters are 0.2714279 0.4950769 2350.968 0.2641291 0.222681 0.8060575 0.4891692
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
```

Going forward, for now, only the Gauss kernel is dealt with; Matern 5/2 will be returned to once I've figured out the estimates returned by the two packages and the derivative of the Matern 5/2 kernel.

1.2 Verifying the estimates by-hand for both estimation methods

To calculate the posterior predictions at the 100 testing points by-hand, we require the matrix H containing the inputs for the 196 observations and well as matrix A, its inverse, and $\mathbf{t}(x^*)$ for all 196 of the new inputs x^*. A and $\mathbf{t}(x^*)$ are built using the estimate of the range parameter returned by each package. By-hand predictions match those returned by the two estimation methods (Figure 3).

Note that for `solve`, which is a numerical way to calculate the inverse of a matrix, if some of the numbers used in the inverse calculation for the matrix `A` are very small, it assumes that they are zero, leading to the assumption that it is a singular matrix. This is why it specifies that they are computationally singular, because the matrix might not be singular given a different tolerance. The solution is to set the tolerance for when it assumes that they are zero. `solve` allows you to specify this parameter (`tol`).

For both we calculate

$$\frac{\sum_{i=1}^{N}\left|(\text{predictions from package})_i - (\text{predictions by-hand})_i\right|}{N},$$

where N is the number of test points, then divide this by 10^{-8} (machine precision) to get 0 for MLE and 0.0002 for MAP.

We create an object containing the emulated partial derivatives for use in the alignment measure. Finally, we create an object containing the normalised partial derivatives for use in the alignment measure.

2 Variable 2: AOD at longitude -169.6875 and latitude 51.875

We start with AOD at the gridbox highlighted in green in Figure 1.

2.1 Emulator fit and validation

We have a 7-dimensional input, one output (AOD in July at the gridbox with longitude -169.6875 and latitude 51.875), and 196 noise-free observations. We will assume a linear mean function of the form

$$m(x) = \beta_0 + \beta_1(\text{bl\_nuc})\ x_1 + \beta_2(\text{ait\_width}) + \cdots + \beta_{37}(\text{a\_ent\_1\_rp})$$

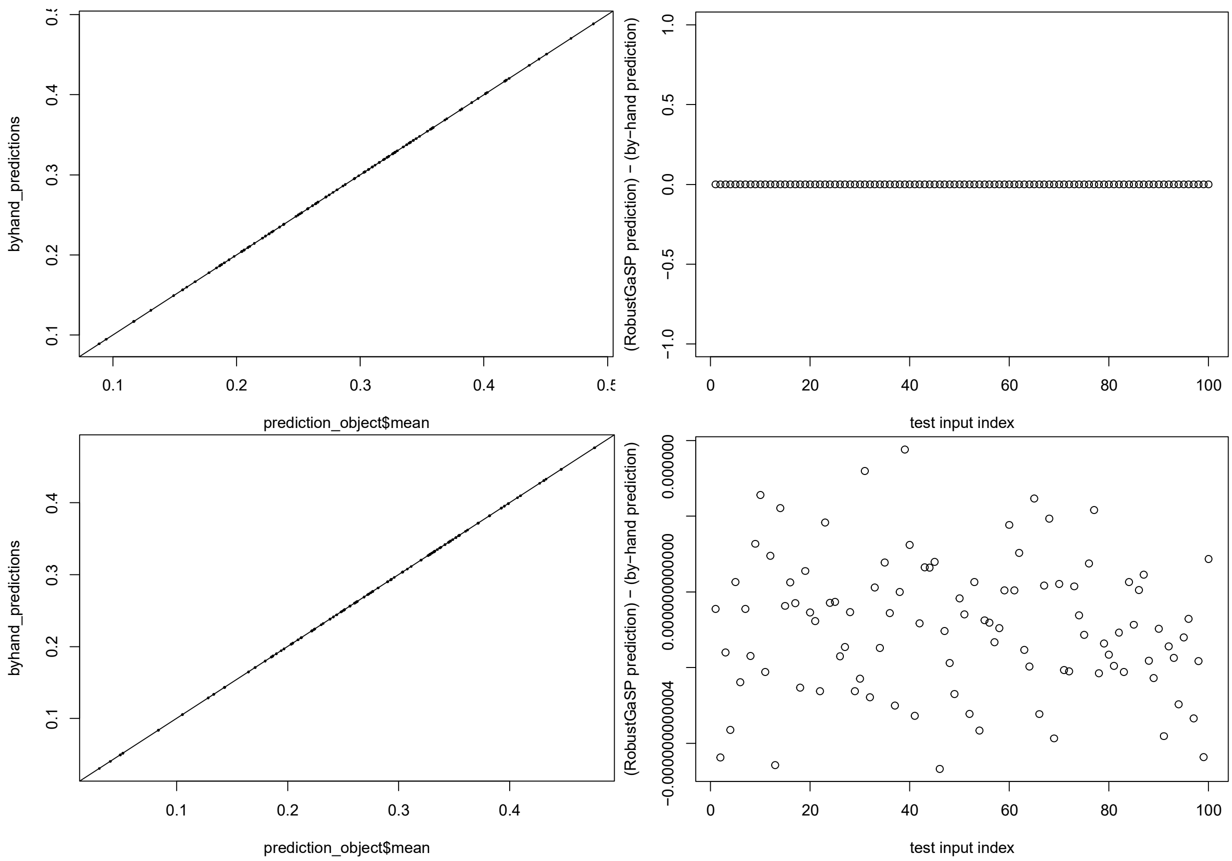


Figure 2: The predictions found by-hand match well with those made by both estimation methods (left), with the latter minus the former found to be essentially zero (right). Top: MLE. Bottom: MAP.

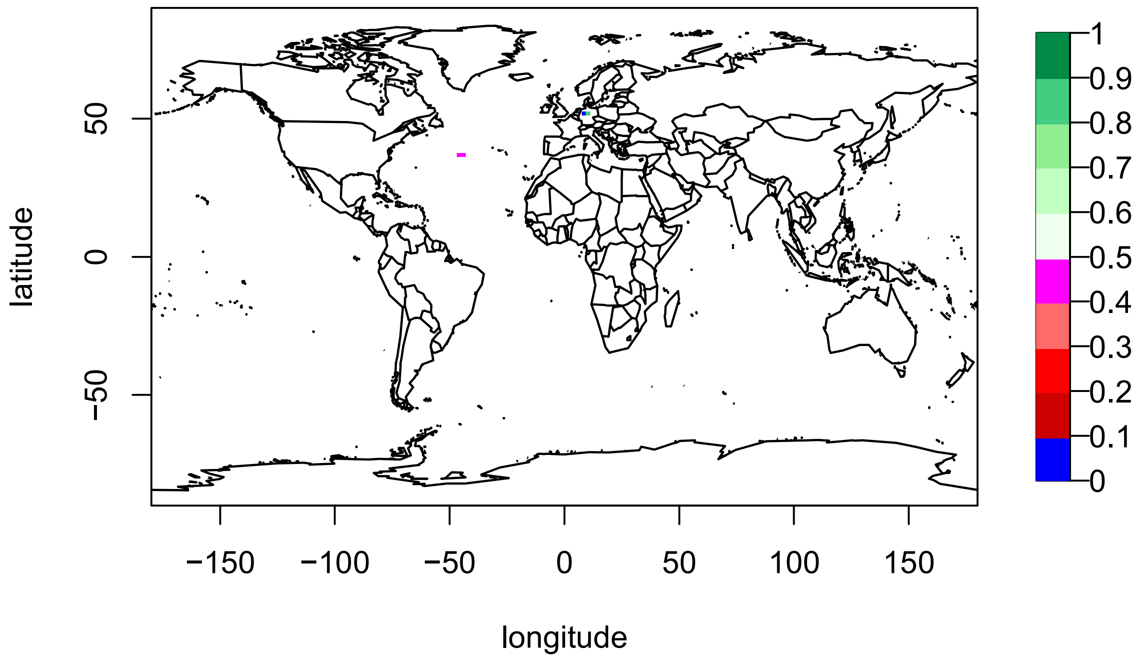


Figure 3: Selected gridbox is highlighted in green

The Gauss (squared exponential) *and* Matern 5/2 kernels will be compared for both estimation methods.

100 new LH points are generated and predictions at these points made. The mean width of the 95% CI's for these predictions are shown in Table 1.

Table 2: Mean width of 95% CI

Estimate method	Kernel	Mean width of 95% CI
MLE	Gauss	0.177
MLE	Matern 5/2	0.177
MAP	Gauss	0.1696
MAP	Matern 5/2	0.1699

Leave-one-out validation is carried out; the results are shown in Figure 2.

```
## The upper bounds of the range parameters are Inf Inf Inf Inf Inf Inf Inf
## The initial values of range parameters are 0.01381671 0.01436526 0.01435065 0.01435916 0.0143668 0.0
## Start of the optimization  1  :
## The number of iterations is  1
##  The value of the  profile likelihood  function is  89.87342
##  Optimized range parameters are 0.01381671 0.01436526 0.01435065 0.01435916 0.0143668 0.01435145 0.0
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 2.760073 3.084807 3.075348 3.080849 3.085803 3.075865 3.0
## Start of the optimization  2  :
## The number of iterations is  2
##  The value of the  profile likelihood  function is  89.87342
##  Optimized range parameters are 0.000000000000002372118 0.0000002811017 0.000183228 0.00000000000107
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The upper bounds of the range parameters are Inf Inf Inf Inf Inf Inf Inf
## The initial values of range parameters are 0.0138241 0.01436526 0.01435065 0.01435916 0.0143668 0.01
## Start of the optimization  1  :
## The number of iterations is  1
##  The value of the  profile likelihood  function is  89.46713
##  Optimized range parameters are 0.0138241 0.01436526 0.01435065 0.01435916 0.0143668 0.01435145 0.01
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 2.764068 3.084807 3.075348 3.080849 3.085803 3.075865 3.0
## Start of the optimization  2  :
## The number of iterations is  2
##  The value of the  profile likelihood  function is  89.46713
##  Optimized range parameters are 0.00000000000004919178 0.00000000002798505 0.000000001088585 0.0000
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The upper bounds of the range parameters are Inf Inf Inf Inf Inf Inf Inf
## The initial values of range parameters are 0.0138241 0.01436526 0.01435065 0.01435916 0.0143668 0.01
## Start of the optimization  1  :
## The number of iterations is  1
##  The value of the  profile likelihood  function is  89.52325
##  Optimized range parameters are 0.0138241 0.01436526 0.01435065 0.01435916 0.0143668 0.01435145 0.01
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
```

```
## The number of iterations is  67
##  The value of the  marginal posterior  function is  4.312316
##  Optimized range parameters are 98269.39 12667894733663108472028802662464820242200008648 10046759755
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 2.764068 3.084807 3.075348 3.080849 3.085803 3.075865 3.0
## Start of the optimization  2  :
## The number of iterations is  55
##  The value of the  marginal posterior  function is  65.9039
##  Optimized range parameters are 0.2590941 0.523236 225.5885 0.2633644 0.2245545 0.8879048 0.4627195
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
```

Going forward, for now, only the Gauss kernel is dealt with; Matern 5/2 will be returned to once I've figured out the estimates returned by the two packages and the derivative of the Matern 5/2 kernel.

2.2 Verifying the estimates by-hand for both estimation methods

To calculate the posterior predictions at the 100 testing points by-hand, we require the matrix H containing the inputs for the 196 observations and well as matrix A, its inverse, and $\mathbf{t}(x^*)$ for all 196 of the new inputs x^*. A and $\mathbf{t}(x^*)$ are built using the estimate of the range parameter returned by each package. By-hand predictions match those returned by the two estimation methods (Figure 3).

Note that for `solve`, which is a numerical way to calculate the inverse of a matrix, if some of the numbers used in the inverse calculation for the matrix `A` are very small, it assumes that they are zero, leading to the assumption that it is a singular matrix. This is why it specifies that they are computationally singular, because the matrix might not be singular given a different tolerance. The solution is to set the tolerance for when it assumes that they are zero. `solve` allows you to specify this parameter (`tol`).

For both we calculate

$$\frac{\sum_{i=1}^{N}\left|(\text{predictions from package})_i - (\text{predictions by-hand})_i\right|}{N},$$

where N is the number of test points, then divide this by 10^{-8} (machine precision) to get 0 for MLE and 0.0002 for MAP.

We create an object containing the emulated partial derivatives for use in the alignment measure. Finally, we create an object containing the normalised partial derivatives for use in the alignment measure.

3 Alignment measure

```
## [1] 0.879338
```

```
## [1] 0.9989766
```

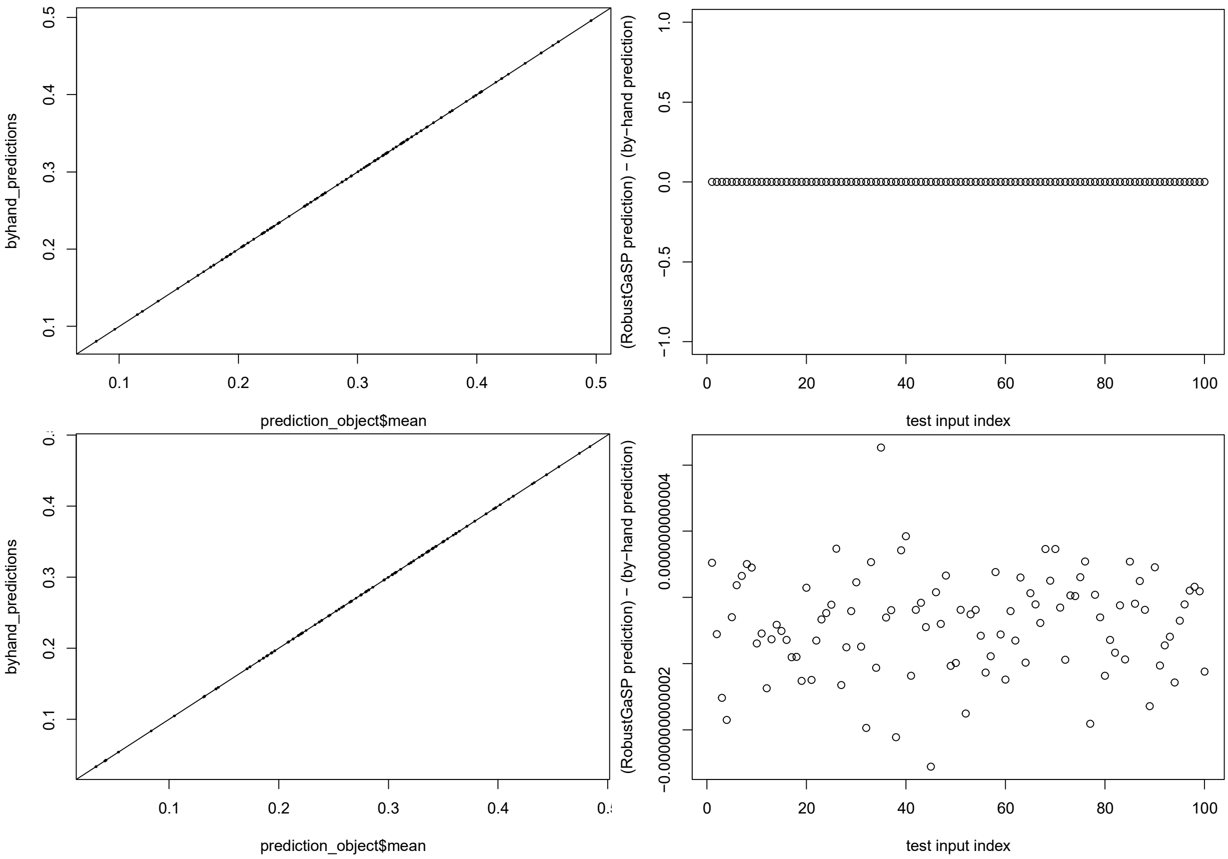


Figure 4: The predictions found by-hand match well with those made by both estimation methods (left), with the latter minus the former found to be essentially zero (right). Top: MLE. Bottom: MAP.